

L6. From Digital to Quantum Circuits

Xin Wan

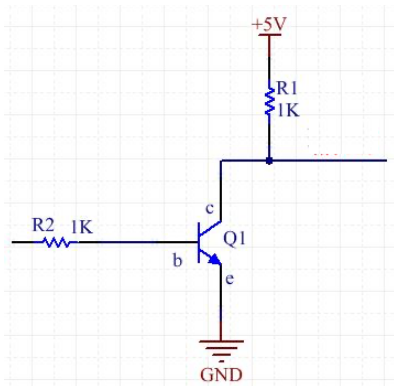
Zhejiang University

Outline

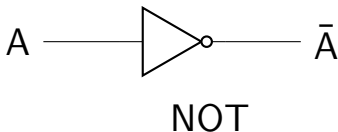
- Digital Electronics
- The No-Cloning Theorem
- Computation with Quantum Gates
- Quantum Teleportation, an Example of Quantum Circuit
- Deferred Measurement

6.1 Digital Electronics

- In digital electronics, we represent bit value by voltages: the logical value 1 is a high voltage (typically +5 V) and 0 is the ground voltage (0 V).
- The voltage bits are coupled and manipulated by transistor-based devices, or gates. The simplest gates act on only one bit or combine two bits to generate a single new bit, the value of which is determined by the input.

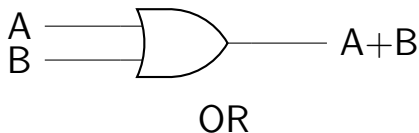
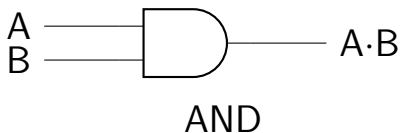


- A gate, or a logical operation, is often represented by a truth table, which presents all of the possible input values and the associated outputs.
- For a single bit, with value 0 or 1, the only possible operations are the trivial identity and the bit flip (NOT gate).



A	$\bar{A}$
0	1
1	0

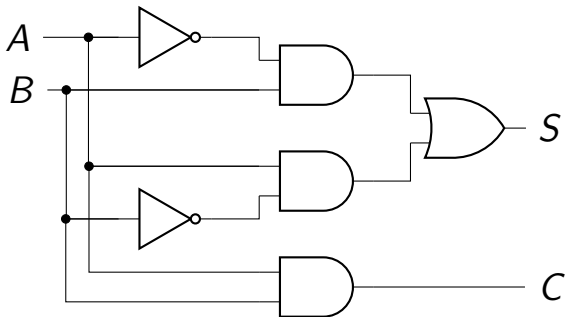
- The simplest two-bit gates are the AND gate and the OR gate.



A	B	$A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

A	B	$A + B$
0	0	0
0	1	1
1	0	1
1	1	1

- The basic gates can be combined to evaluate any possible Boolean function. A classical example is the half-adder, which is a primitive for performing binary addition. Which operations do you think quantum computing cannot do?



A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



I think I can safely say that nobody understands quantum mechanics.

(Richard Feynman)

izquotes.com

6.2 No-Cloning Theorem: Possible to Copy?

- Let us begin with a discussion on to what extent can we duplicate the information of a quantum state.
 - The first possibility is to copy or clone, as if we had a quantum copy machine. Putting in a qubit in a state $|\psi\rangle = \mu|0\rangle + \nu|1\rangle$, we would get a new qubit in the state $|\psi\rangle = \mu|0\rangle + \nu|1\rangle$ and the original qubit in the state $|\psi\rangle$ back. In other words, the result is

$$|\psi\rangle|\psi\rangle = \mu^2|0\rangle|0\rangle + \mu\nu|0\rangle|1\rangle + \nu\mu|1\rangle|0\rangle + \nu^2|1\rangle|1\rangle.$$

- The second possibility is to entangle, which we have learned in Lecture 4. Given a quantum state $|\psi\rangle = \mu|0\rangle + \nu|1\rangle$ and another qubit, say, in the state $|0\rangle$, we obtain $\mu|0\rangle|0\rangle + \nu|1\rangle|1\rangle$.

The No-Cloning Theorem

It is not possible to perfectly clone an unknown quantum state, or a state drawn from a set of two (or more) non-orthogonal states. [Wooters & Zurek (1982); Dieks (1982)]

Proof by Contradiction

- Assume an ideal cloner exists, which is described by a unitary operator U_c that acts on the global system of the initial copy in the pure state $|\psi\rangle$, a blank copy on which the state will be cloned, say in $|0\rangle$, and in general an auxiliary system (ancilla) in a state $|a_0\rangle$.

- Assume that ideal cloning is possible for nonorthogonal input state $|\psi\rangle$ and $|\phi\rangle$. The cloning transformation would lead to

$$U_c(|\psi\rangle|0\rangle|a_0\rangle) = |\psi\rangle|\psi\rangle|a_\psi\rangle,$$

$$U_c(|\phi\rangle|0\rangle|a_0\rangle) = |\phi\rangle|\phi\rangle|a_\phi\rangle.$$

where $|a_\psi\rangle$ and $|a_\phi\rangle$ are the output states of the ancilla.

- The inner product of the two equations yields

$$\langle\psi|\phi\rangle = (\langle\psi|\phi\rangle)^2 \langle a_\psi|a_\phi\rangle.$$

- We obtain

$$\langle \psi | \phi \rangle = \frac{1}{\langle a_\psi | a_\phi \rangle}.$$

This can be satisfied only when $\langle \psi | \phi \rangle = 1$, because $|\langle a_\psi | a_\phi \rangle| \leq 1$.

- Therefore, U_c does not exist. Only approximate cloning transformations can exist.
- The no-cloning theorem is crucial in the security of quantum key distribution, during which Eve cannot clone the encoded photons and measure them later.

6.3 Computation with Quantum Gates

- A computation is a map, $f : x \rightarrow f(x)$, where x is an n -bit integer, and $f(x)$ an m -bit integer.
- In a quantum computer, the total resources are, at least, $n + m$ qubits. The computation is achieved by applying a unitary transformation U_f to the $n + m$ qubits.

$$U_f(|x\rangle_n |y\rangle_m) = |x\rangle_n |y \oplus f(x)\rangle_m,$$

where $\oplus$ represents the bitwise exclusive OR, or mod-2 addition. For example,

$$110010 \oplus 010101 = 100111.$$

Reversibility

- The computation is reversible.

$$\begin{aligned}U_f U_f(|x\rangle_n |y\rangle_m) &= U_f(|x\rangle_n |y \oplus f(x)\rangle_m) \\&= |x\rangle_n |y \oplus \underbrace{f(x) \oplus f(x)}_{z \oplus z = 0, \forall z}\rangle_m \\&= |x\rangle_n |y\rangle_m\end{aligned}$$

- In particular, when $y = 0$,

$$U_f(|x\rangle_n |0\rangle_m) = |x\rangle_n |f(x)\rangle_m.$$

Quantum Gates

- Therefore, a quantum gate is a *reversible* operation that a quantum computer can perform upon qubits.
- A gate is a linear transformation on the state of the qubit that takes unit vectors into unit vectors. Such a transformations u is called **unitary** and satisfies the condition

$$uu^+ = u^+u = 1.$$

- In designing quantum algorithms, the class of allowed unitary transformations is almost always restricted to a small (but universal) set of single-qubit gates and two-qubit gates.

- We begin with single-qubit gates. Note that the NOT operation X is the only nontrivial reversible operation a classical computer can perform on a single bit.
- NOT gate (Pauli- X matrix):

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

- Examples:

$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

- NOT gate flips the qubit, just like a classical bit.

- Hadamard gate can flip the qubit half way, such that we have a *superposition* of 0 and 1.

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

- Roughly speaking, $H \sim \sqrt{\text{NOT}}$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

- H-gates are often used to implement quantum parallelism.

- We can also shift the phase of the qubit.
 - The Pauli-Z operator rotates the state about the z-axis by 180° .

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- The S operator rotates the state about the z-axis by 90° .

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

- The T operator rotates the state about the z-axis by 45° .

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

Quantum Parallelism

- Let us illustrate why, in principle, a quantum computer can achieve superior performance to their classical counterpart.

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle),$$

$$\begin{aligned}(H \otimes H)(|0\rangle \otimes |0\rangle) &= (H|0\rangle) \otimes (H|0\rangle) \\ &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ &= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle).\end{aligned}$$

- In general,

$$H^{\otimes n}|0\rangle_n = \frac{1}{2^{n/2}} \sum_{0 \leq x < 2^n} |x\rangle_n,$$

where $H^{\otimes n} = \underbrace{H \otimes H \otimes \dots \otimes H}_{n \text{ times}}.$

- Therefore, we seems to calculate in parallel universes.

$$\begin{aligned} U_f(H^{\otimes n} \otimes I_m)(|0\rangle_n|0\rangle_m) &= \frac{1}{2^{n/2}} \sum_{0 \leq x < 2^n} U_f(|x\rangle_n|0\rangle_m) \\ &= \frac{1}{2^{n/2}} \sum_{0 \leq x < 2^n} |x\rangle_n |f(x)\rangle_m \end{aligned}$$

- The quantum advantage is that the (unknown) state encodes the possibilities of extracting the information of $f(x)$ for all x . But how do we find out what the state is? The not-so-obvious disadvantage is that we cannot duplicate the information of an unknown quantum state.
- Fortunately, it is sufficient to show that a quantum computer can perform tricks, though in very limited cases, that no classical computer can accomplish. Examples include Deutsch's algorithm and Grover's search algorithm, which we will demonstrate in a few weeks.

Two-Qubit Gates

- The quantum controlled-NOT gate is a two-qubit gate.
 - If the state of the i th qubit (the control qubit) is $|0\rangle$, C_{ij} leaves the state of the j th qubit (the target qubit) unchanged.
 - If the state of the control qubit is $|1\rangle$, C_{ij} applies the NOT operator X to the state of the target qubit.
- Example:

$$\begin{aligned} & C_{12}(a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle) \\ &= a|00\rangle + b|01\rangle + c|11\rangle + d|10\rangle \end{aligned}$$

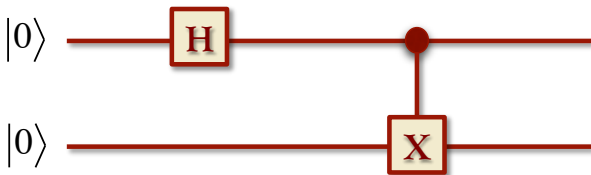
- CNOT gate can be used to entangle two qubits. Note that

$$a|00\rangle + c|10\rangle = (a|0\rangle + c|1\rangle)|0\rangle.$$

- Matrix representation

$$C_{12} = \text{CNOT} \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad C_{21} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

- A quantum circuit: What is the final state?



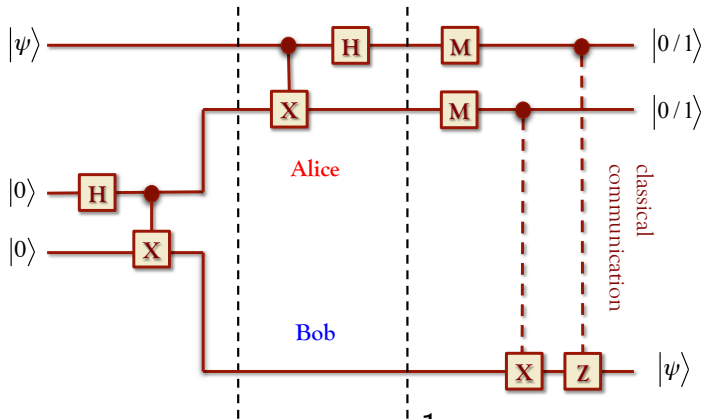
Universality of Quantum Gates

- A small set of gates (e.g. AND, OR, NOT) can be used to compute an arbitrary classical function. We say that such a set of gates is universal for classical computation.
- Similar universality result is true for quantum computation, where a set of gates is said to be *universal* for quantum computation if any unitary operation may be approximated to arbitrary accuracy by a quantum circuit involving only those gates. However, *approximating arbitrary unitary gates is generically hard*.
- The standard set of universal gates consists of the Hadamard (H), phase (S), $\pi/8$ (T) and CNOT gates.

6.4 Quantum Teleportation

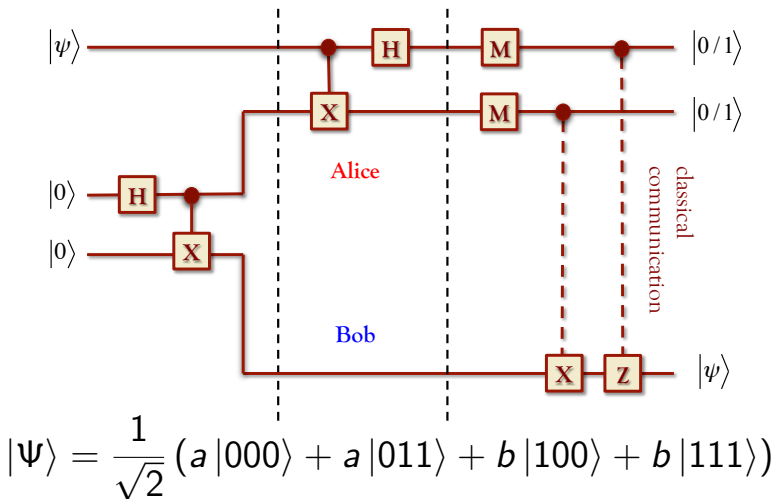
- Teleportation is not the transfer of an object by disappearing it from one point in space and reappearing it in another location.
- It is a transfer of the information that fully characterizes all properties of the object.
- Suppose Alice has a qubit in an unknown state $|\psi\rangle_A = a|0\rangle + b|1\rangle$. In teleportation, her goal is to transfer the information (i.e. a and b), not the physical qubit, to Bob at a different location.

- After Stage 1:

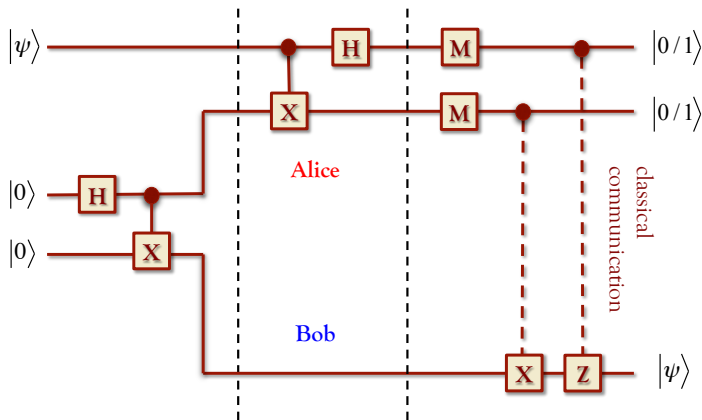


$$|\Psi\rangle = (a|0\rangle + b|1\rangle) \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- Equivalently,

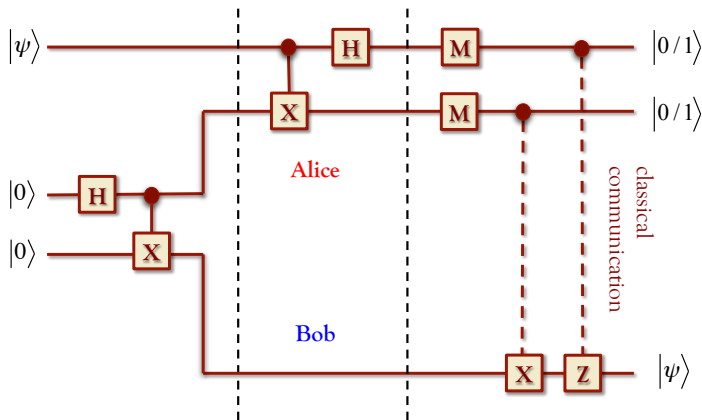


- After Stage 2:



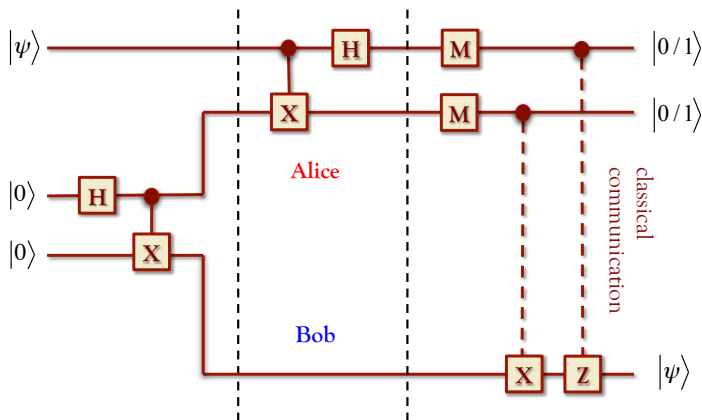
$$|\Psi\rangle = \frac{1}{2} (a|000\rangle + a|100\rangle + a|011\rangle + a|111\rangle + b|010\rangle - b|110\rangle + b|001\rangle - b|101\rangle)$$

- Equivalently,

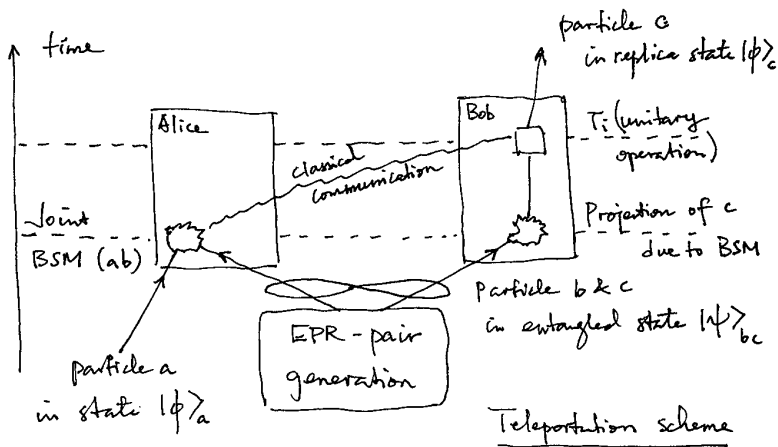


$$|\Psi\rangle = \frac{1}{2} [|00\rangle (a|0\rangle + b|1\rangle) + |01\rangle (a|1\rangle + b|0\rangle) + |10\rangle (a|0\rangle - b|1\rangle) + |11\rangle (a|1\rangle - b|0\rangle)]$$

- Equivalently,



$$|\Psi\rangle = \frac{1}{2} [|00\rangle I + |01\rangle X + |10\rangle Z + |11\rangle XZ] (a|0\rangle + b|1\rangle)$$



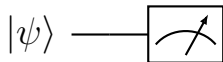
- Is it a violation of the no-cloning theorem? Or locality?

- Alice has no knowledge of the quantum state, even though she is able to teleport it.
- Nor does Alice maintain a copy of the quantum state. After the teleportation, the original qubit ends up either in $|0\rangle$ or $|1\rangle$, depending on the measurement result.
- The classical communication channel is limited by the speed of light, so quantum teleportation cannot be accomplished faster than the speed of light.

Experiments: Bouwmeester et al., Nature 390, 575 (1997); Kim et al., Phys. Rev. Lett. 86, 1370 (2001); Ursin et al., Nature 430, 849 (2004).

6.5 Deferred Measurement

- In the teleportation circuit, we have projective measurements in the computational basis denoted by M . Alternatively, we can use a meter symbol

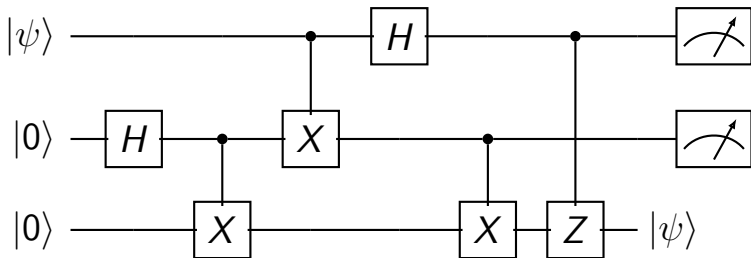


- In the theory of quantum circuits it is conventional to not use any special symbols to denote more general measurements, because, as will be discussed later, they can always be represented by unitary transforms with ancilla qubits followed by projective measurements.

- As in the teleportation circuit, quantum measurements in a circuit are often performed at an intermediate step, and the measurement results are used to conditionally control subsequent quantum gates.
- However, such measurements can always be moved to the end of the circuit. Classically conditioned operations can be replaced by quantum conditioned operations.
 - Of course, the interpretation of this circuit as performing teleportation is lost, because no classical information is transmitted from Alice to Bob, but the overall action of the two quantum circuits is the same, which is the key point.

Principle of deferred measurement

Measurements can always be moved from an intermediate stage of a quantum circuit to the end of the circuit; if the measurement results are used at any stage of the circuit then the classically controlled operations can be replaced by conditional quantum operations.



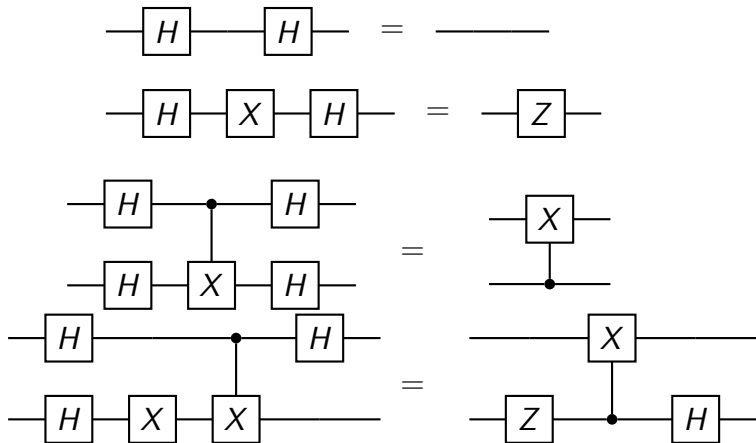
- Measurement is generally considered to be an irreversible operation, destroying quantum information and replacing it with classical information.
- In certain carefully designed cases, however, this need not be true, as illustrated by teleportation. In teleportation the measurement result does not reveal any information about the identity of the quantum state being teleported.
 - In order for a measurement to be reversible, it must reveal no information about the quantum state!
 - The principle turns out to be crucial in quantum error correction.

References:

- N. David Mermin, Quantum Computer Science, An Introduction (Cambridge University Press, 2007), Chap. 1.
- Stephen M. Barnett, Quantum Information (Oxford University Press, 2009), Chap. 6.
- Jack D. Hidary, Quantum Computing: An Applied Approach, 2nd ed. (Springer, 2021); download at <https://link.springer.com/book/10.1007/978-3-030-83274-2>.

Homework

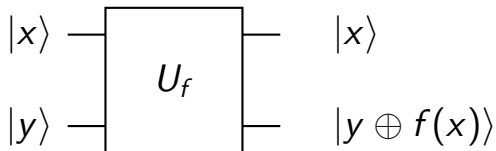
P6-1. Show that the following pairs of circuits are equivalent.



P6-2. Assume a computation is achieved by applying a unitary transformation U_f to $n + m$ qubits.

$$U_f(|x\rangle_n |y\rangle_m) = |x\rangle_n |y \oplus f(x)\rangle_m,$$

where $\oplus$ represents the bitwise exclusive OR, or mod-2 addition, e.g., $110010 \oplus 010101 = 100111$. This can be illustrated by a quantum circuit, say, for $n = m = 1$.



Show that one can raise the function $f(x)$ to an exponential form through the following trick:

$$U_f(|x\rangle \otimes H|1\rangle) = (-1)^{f(x)} |x\rangle \otimes H|1\rangle.$$

Note that the application of U_f on the output qubit y can be regarded as the identity, so it is never entangled with the input qubits and remains a constant state.

P6-3. Superdense coding is an application of entangled quantum states. The protocol involves two parties, conventionally known as Alice and Bob. Their goal is to transmit two bits of classical information from Alice and Bob by sending only one qubit.

Suppose Alice and Bob initially share a pair of qubits in the entangled state

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

If Alice wants to send classical information $|00\rangle$ to Bob, she does nothing to her qubit. If she wants to send $|01\rangle$, $|10\rangle$, or $|11\rangle$ to Bob, she applies Z , X , or ZX to her qubit. What are the results of the three operations?

Alice then sends her qubit to Bob, who can make joint measurement to both qubits. How shall Bob measure the qubits and, thus, acquire the information Alice wants to send? In other words, what measurements can Bob apply to distinguish the four possible orthogonal states?